Ex93_Adaptive_line_enhancer

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1 MSE StatDig : Chap 9 “Adaptive filtering””

1.1 Ex 9.3 Adaptive line enhancer using LMS

ver : DLY/12.05.2025

1.2 General

1.2.1 Description

This example shows how to apply adaptive filters to signal separation. In adaptive line enhancement, a measured signal s(n) contains two signals, an unknown signal of interest v(n), and a nearly-periodic noise signal p(n). The goal is to remove the noise signal p(n) from the measured signal s(n) to obtain the signal of interest v(n).

1.2.2 Work

Ex1 : Load the signal

- Load from the file ex93_AdaptiveLineEnhancer_LMS.wav. The file contains the data x and the frequency sampling Fe.

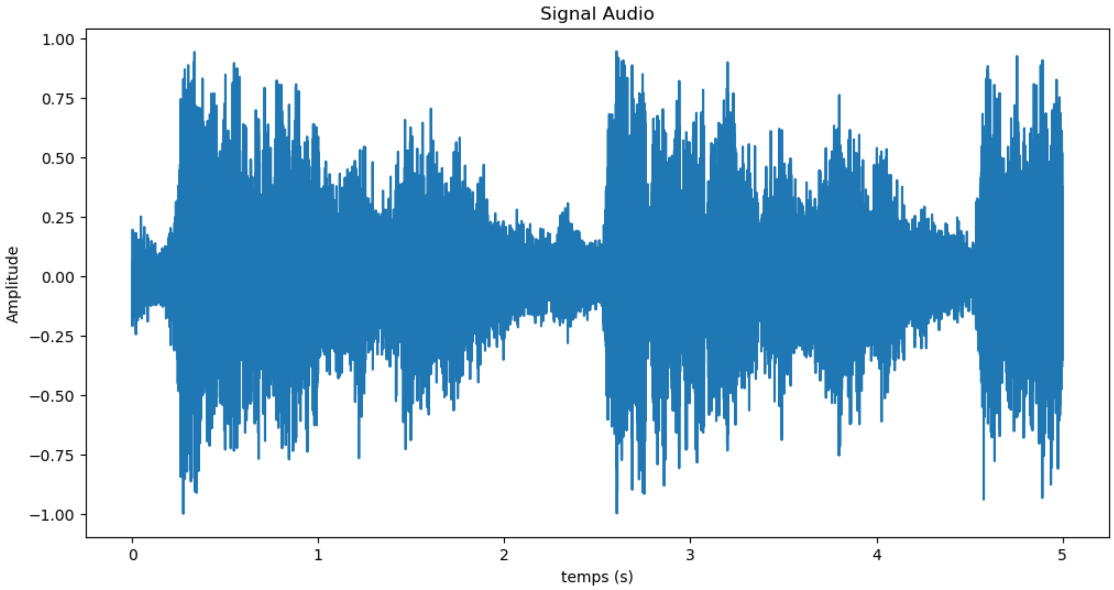
```
from scipy.io import wavfile
# Charger le fichier WAV
fs, signal = wavfile.read('ex93_AdaptiveLineEnhancer_LMS.wav')
```

- Normalize the signal to the range [-1, 1]

```
signal = signal / max(abs(signal))
```

- Create a shorter version which will be faster to compute. Example tmax = 3 seconds instead of 9 seconds of the original signal
- Hear the signal with sounddevice and display it with matplotlib

```
import sounddevice as sd
sd.play(signal_cut, fs)
sd.wait()  # Attendre que la lecture soit terminée
```

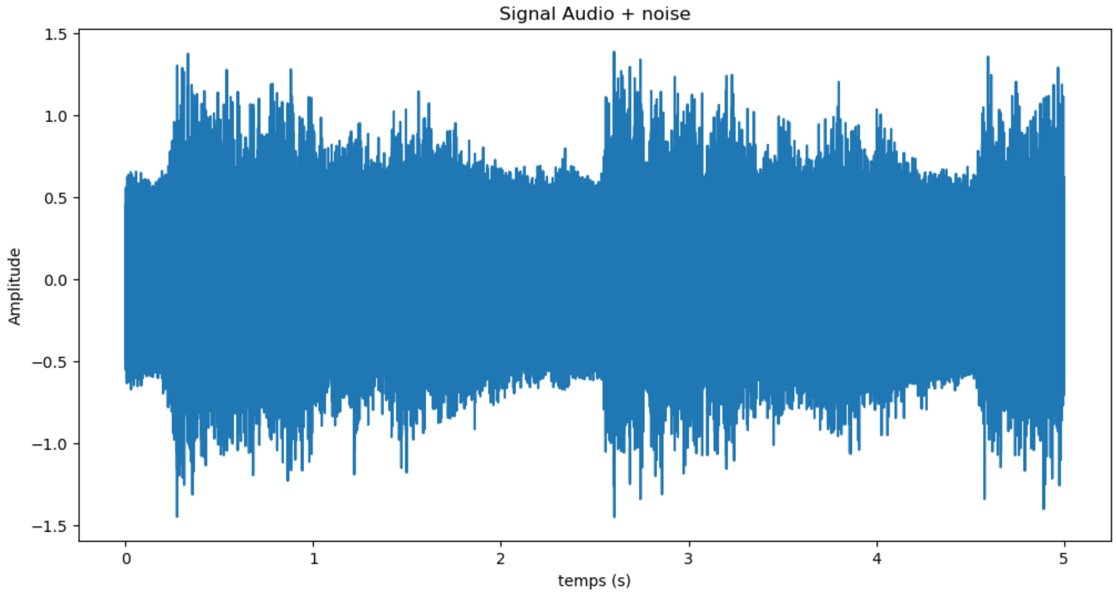


Ex2 : One frequency noise The goal now is to add a periodic perturbation to the signal and to remove it with two methods : notch filter and lms adaptive filter

Ex2.1 : Generate the noise signal

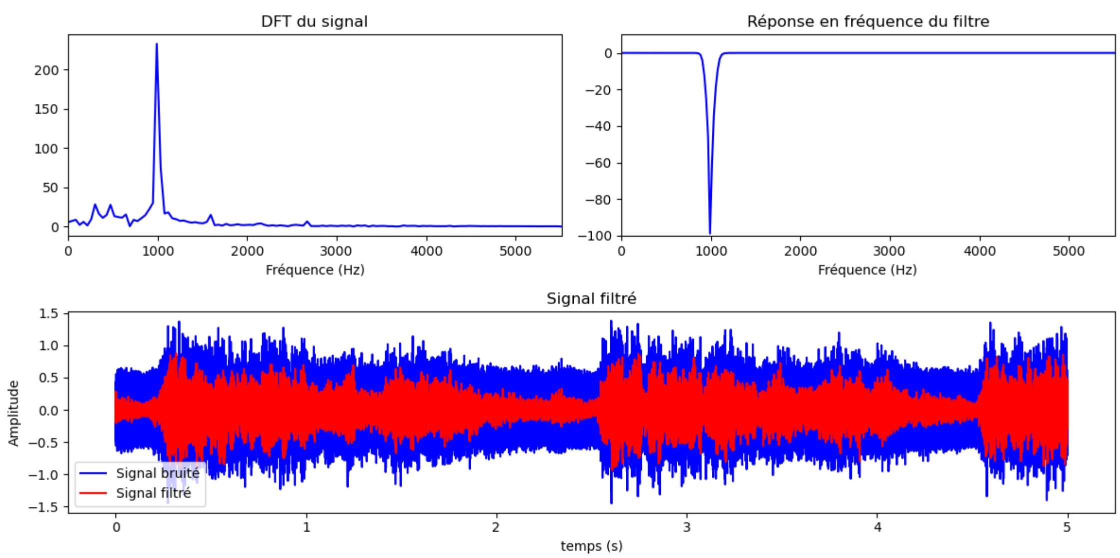
- Generate a periodic noise signal p(n), a sinusoid with a frequency of 1000 Hz and an amplitude of 0.5
- Add this noise p(n) to the signal x(n) to create the measured signal s(n)
- Hear the noise and the noisy signal with sounddevice

```
Text(0, 0.5, 'Amplitude')
```



Ex2.2 : Stopband filter As the noise p(n) is a single tone, it is very easy to remove it with a stopband filter.

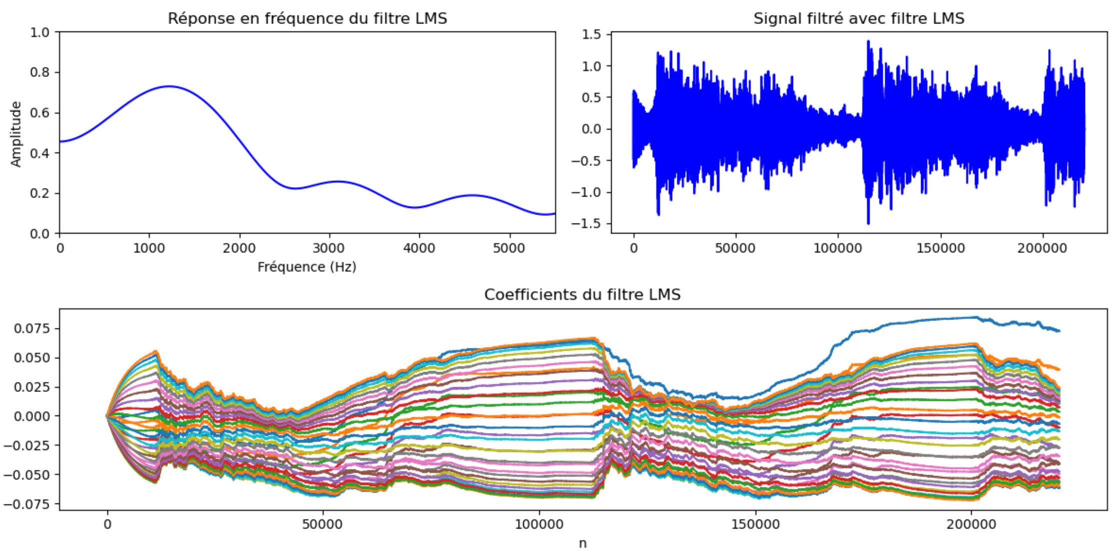
- Analyse the noisy signal in the frequency domain
- Design a stopband filter to remove the noise
- Filter the signal with the notch filter and display it and hear the filtered signal



Ex3 : LMS noise filtering An adaptive line enhancer (ALE) is based on the straightforward concept of linear prediction. A nearly-periodic signal can be perfectly predicted using linear combinations of its past samples, whereas a non-periodic signal cannot. So, a delayed version of the measured signal s(n-D) is used as the reference input signal x(n) to the adaptive filter, and the desired response signal d(n) is made equal to s(n). The parameters to choose in such a system are the signal delay D and the filter length L used in the adaptive linear estimate. The amount of delay depends on the amount of correlation in the signal of interest. Since we don't have this signal, we shall just pick a value of D=100 and vary it later. Such a choice suggests that samples of the Hallelujah Chorus are uncorrelated if they are more than about 12 msec apart. Also, we'll choose a value of L=32 for the adaptive filter, although this too could be changed.

3.1 : Design filter

- Select as input of the LMS the signal x delayed by Delay = 100 and the desired output the signal not delayed (x =xn(1:end-Delay), d = xn(Delay:end))
- Choose a LMS with 32 coefficients and a
- Reconstruct the signal x by removing the estimated signal obtained with LMS filter

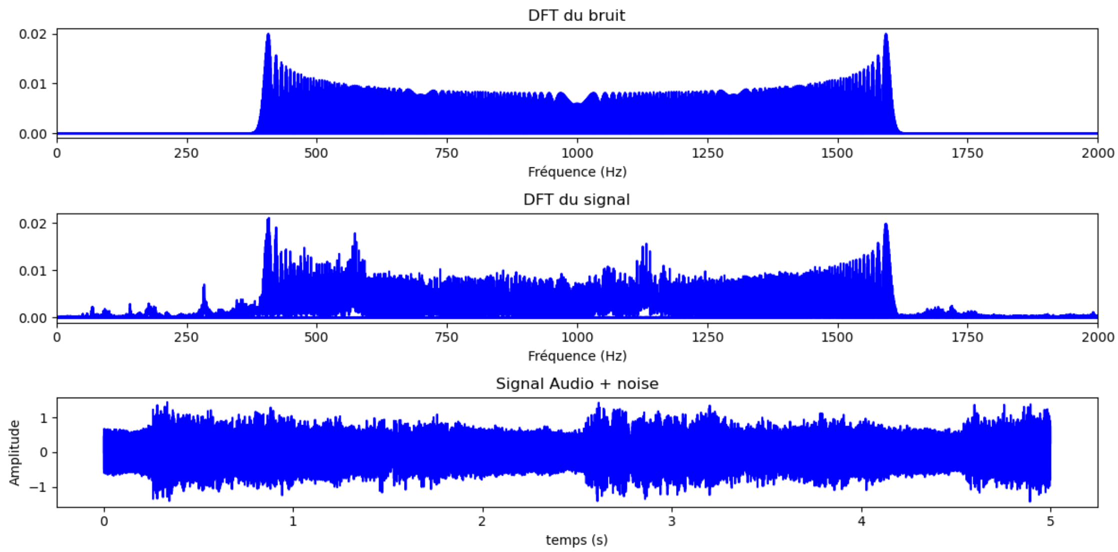


Ex4 : FM Noise source As seen, removing a pure sinusoid from a sinusoid plus music signal is not particularly challenging if the frequency of the offending sinusoid is known. So, let's make the problem a bit harder by adding an FM-modulated sinusoidal signal as our noise source.

$$\text{noise}_{\text{fm}} = A \cdot \sin\left(2\pi f_0 t + \beta \sin(2\pi f_m t)\right)$$

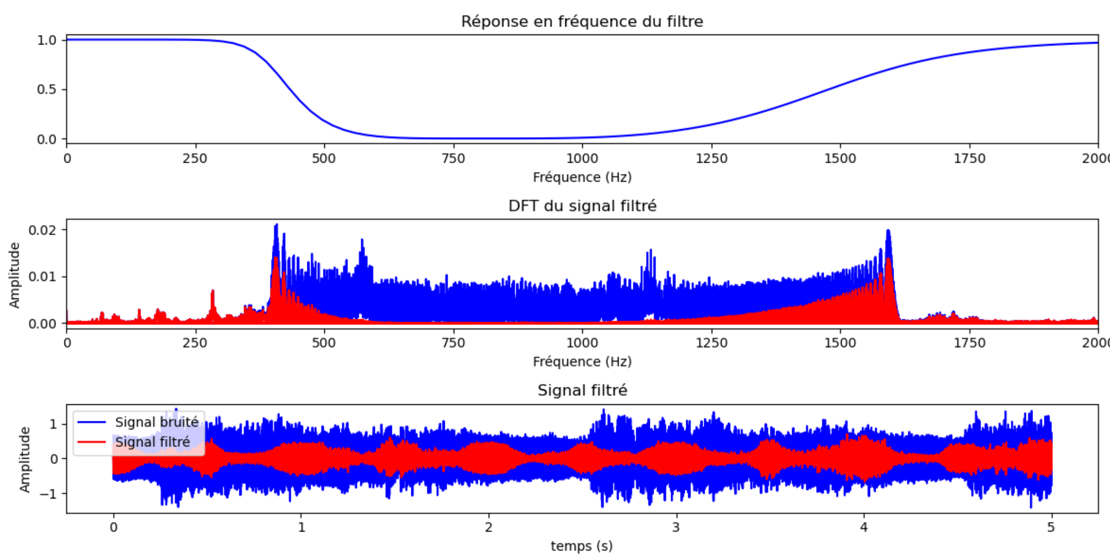
4.1 : Generate the FM noise

- Generate a noise pfm(n) signal which is a FM-modulated sinusoidal with A = 0.5, f0=1000Hz, fm=1Hz, $\beta = 600$
- Add this noise pfm(n) to the signal x(n) to create the new noisy signal sfm(n)
- Hear the signal



4.2 : Stop band filter

- Analyse the perturbation in the frequency domain
- Design a stopband filter with $fc = [f_0 - \beta, f_0 + \beta]$ Hz
- Filter the signal with the stopband filter, display it and hear the filtered signal



4.3 : LMS noise filtering

- Reconstruct the signal x by removing the estimated signal obtained with LMS filter ($\mu = 0.01$, p=32, Delay=100)
- Try to change the parameters of the LMS filter to see if you can improve the result

- Display the result and hear the filtered signal